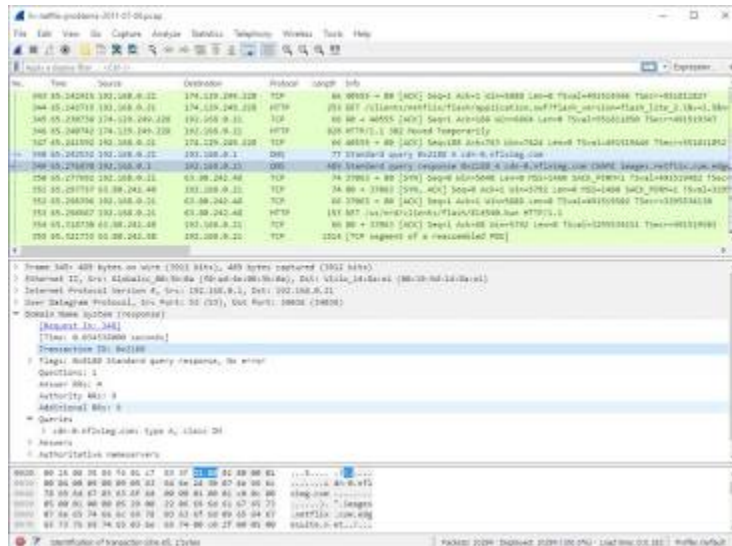


DAY 1 OF INTRODUCTION OF NETWORKING & OSI LAYERS

Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like [instagram.com](https://www.instagram.com)). Stop the capture and note down the protocol names you see in the packet list.

ANS:



Here are the core protocol names you will see in Wireshark when visiting [instagram.com](https://www.instagram.com):

- **DNS:** Resolves [instagram.com](https://www.instagram.com) to an IP address.
- **TCP / UDP:** Sets up the connection (TCP handshake or UDP for QUIC).
- **TLS:** Handles the secure encryption handshake (TLSv1.3).
- **HTTP/2 or HTTP/3:** Application traffic flowing inside the secure tunnel.

Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

ANS:

Here are 3 common packets from a web traffic capture mapped to their respective OSI layers:

1. **DNS (Domain Name System)**
 - **OSI Layer:** Application Layer (Layer 7)

- **In Wireshark:** Appears as standard query/response packets used to translate domain names like `instagram.com` into numeric IP addresses before a connection is made.
2. **TCP (Transmission Control Protocol)**
- **OSI Layer:** Transport Layer (Layer 4)
 - **In Wireshark:** Appears during connection setup (`SYN`, `ACK`) or data transmission, ensuring reliable, ordered, and error-checked delivery between your device and the server.
3. **IP (Internet Protocol / IPv4 or IPv6)**
- **OSI Layer:** Network Layer (Layer 3)
 - **In Wireshark:** Found wrapped around transport and application data, handling logical source and destination addressing to route packets across separate networks.

Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

ANS:



OSI Layers & Daily App Examples

1. Layer 7: Application Layer

- **What it does:** Interfaces directly with software applications that interact with network services.
 - **Daily App Example: WhatsApp or Instagram** (sending a message or loading your feed).
2. **Layer 6: Presentation Layer**
- **What it does:** Translates, encrypts, and compresses data so the application can read it (handles formats, SSL/TLS encryption).
 - **Daily App Example: Instagram** formatting and decompressing compressed image formats (like JPEG/HEIC) to display photos on your screen.
3. **Layer 5: Session Layer**
- **What it does:** Manages sessions and dialogues (opening, maintaining, and terminating connections) between computers.
 - **Daily App Example: Spotify** maintaining an active audio-streaming session without logging you out or losing sync while music plays.
4. **Layer 4: Transport Layer**
- **What it does:** Ensures reliable, ordered data delivery and error checking (using TCP or UDP).
 - **Daily App Example: Zoom or FaceTime** using UDP for fast, real-time video streaming where slight dropped packets are preferred over lag.
5. **Layer 3: Network Layer**
- **What it does:** Handles logical addressing and routing packets across different networks (IP addresses).
 - **Daily App Example: Google Maps** routing traffic packets across global routers to pinpoint your exact GPS location.
6. **Layer 2: Data Link Layer**
- **What it does:** Handles node-to-node data transfer across the local physical network (uses MAC addresses).
 - **Daily App Example:** Your phone communicating directly with your home Wi-Fi router or Bluetooth speaker via MAC addresses.
7. **Layer 1: Physical Layer**
- **What it does:** Transmits raw binary bits (0s and 1s) as electrical voltages, radio waves, or light pulses across physical media.
 - **Daily App Example:** Wi-Fi radio waves transmitting data through the air from your phone to your wireless access point.

Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.

ANS:

Here is how a WhatsApp message travels through 4 layers from top to bottom:

1. **Application Layer (Layer 7):** You type "Hey!" in WhatsApp, and the app packages your text and recipient info into application-level data.
2. **Transport Layer (Layer 4):** TCP takes the message, ensures reliable delivery, breaks it into segments, and adds port numbers.
3. **Network Layer (Layer 3):** Encapsulates the segments into packets and adds source/destination IP addresses so routers can guide it across the internet.
4. **Data Link Layer (Layer 2):** Wraps the packets into frames and attaches physical MAC addresses to move the data across your local Wi-Fi or cellular network.